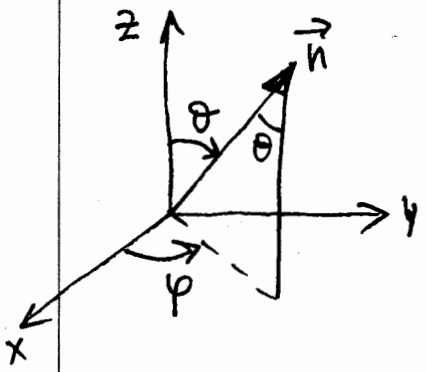
**P3.2** The "standard" solution:

$$\hat{S}_n = \hat{S}\cdot\vec{n} = S_x \sin\theta\cos\varphi + S_y \sin\theta\sin\varphi + S_z\cos\theta$$

$$= \frac{\hbar}{2}\left(\sigma_x \sin\theta\cos\varphi + \sigma_y \sin\theta\sin\varphi + \sigma_z\cos\theta\right)$$




$$= \frac{\hbar}{2}\left[\begin{pmatrix} 0 & \sin\theta\cos\varphi \\ \sin\theta\cos\varphi & 0 \end{pmatrix} + \begin{pmatrix} 0 & -i\sin\theta\sin\varphi \\ i\sin\theta\sin\varphi & 0 \end{pmatrix} + \begin{pmatrix} \cos\theta & 0 \\ 0 & -\cos\theta \end{pmatrix}\right]$$

$$= \frac{\hbar}{2}\begin{pmatrix} \cos\theta & \sin\theta(\cos\varphi - i\sin\varphi) \\ \sin\theta(\cos\varphi + i\sin\varphi) & -\cos\theta \end{pmatrix} =$$

$$= \frac{\hbar}{2}\begin{pmatrix} \cos\theta & e^{-i\varphi}\sin\theta \\ e^{i\varphi}\sin\theta & -\cos\theta \end{pmatrix}$$

$\Rightarrow$ the eigenvalue problem reads

$$\frac{\hbar}{2}\begin{pmatrix} \cos\theta & e^{-i\varphi}\sin\theta \\ e^{i\varphi}\sin\theta & -\cos\theta \end{pmatrix}\begin{pmatrix} \langle +z|\mu\rangle \\ \langle -z|\mu\rangle \end{pmatrix} = \mu\frac{\hbar}{2}\begin{pmatrix} \langle +z|\mu\rangle \\ \langle -z|\mu\rangle \end{pmatrix}$$

where state $|\mu\rangle \equiv |\hat{S}_n = \mu\frac{\hbar}{2}\rangle$ denotes a state with projections $\mu\frac{\hbar}{2}$ on $\vec{n}$.

Dividing by $\frac{\hbar}{2}$:

$$\begin{pmatrix} -\mu + \cos\theta & e^{-i\varphi}\sin\theta \\ e^{i\varphi}\sin\theta & -\mu - \cos\theta \end{pmatrix}\begin{pmatrix} \langle +z|\mu\rangle \\ \langle -z|\mu\rangle \end{pmatrix} = 0$$

Solution exists if determinant of the matrix $= 0$

$$\Rightarrow (\mu - \cos\theta)(\mu + \cos\theta) - e^{-i\varphi}\sin\theta \cdot e^{i\varphi}\sin\theta = 0$$

$$\Rightarrow \mu^2 - \cos^2\theta - \sin^2\theta = 0 \Rightarrow \mu^2 = 1 \Rightarrow \underline{\mu = \pm 1}$$

Start with $\mu=+1$ and denote, for brevity, $a=\langle +z|+1\rangle$
$b=\langle -z|+1\rangle$

$$\Rightarrow \begin{cases} (\cos\theta-1)\,a + e^{-i\varphi}\sin\theta\, b = 0 & (1) \\ e^{i\varphi}\sin\theta\, a - (1+\cos\theta)\, b = 0 & (2) \end{cases}$$

Use identities $\left[\begin{array}{l} \cos\theta = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2} = 2\cos^2\frac{\theta}{2} - 1 = 1 - 2\sin^2\frac{\theta}{2} \\ \sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2} \end{array}\right.$

$$\Rightarrow (1)\quad e^{-i\varphi}\, 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\, b = 2\sin^2\frac{\theta}{2}\, a$$

$$e^{-i\varphi}\cos\frac{\theta}{2}\, b = \sin\frac{\theta}{2}\, a$$

Normalization $\langle +1|+1\rangle = 1$ requires $(a^*, b^*)\begin{pmatrix} a \\ b \end{pmatrix} = |a|^2 + |b|^2 = 1$

$\Rightarrow$ we can choose $b = e^{i\varphi}\sin\frac{\theta}{2}$

$\Rightarrow$ from (1) $a = \cos\frac{\theta}{2}$

$$\Rightarrow |+1\rangle = \cos\frac{\theta}{2}\,|+z\rangle + e^{i\varphi}\sin\frac{\theta}{2}\,|-z\rangle$$

Note that eq. (2) is satisfied by this choice as well:

$$(2)\quad e^{i\varphi}\sin\theta\; a = (1+\cos\theta)\, b$$

$$e^{i\varphi}\, 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\; a = 2\cos^2\frac{\theta}{2}\; b$$

$$e^{i\varphi}\sin\frac{\theta}{2}\; a = \cos\frac{\theta}{2}\; b$$

$$e^{i\varphi}\sin\frac{\theta}{2}\cos\frac{\theta}{2} = \cos\frac{\theta}{2}\, e^{i\varphi}\sin\frac{\theta}{2} \qquad \underline{\underline{\text{OK}}}$$

Next $\mu = -1$

$$\Rightarrow \begin{pmatrix} 1+\cos\theta & e^{-i\varphi}\sin\theta \\ e^{i\varphi}\sin\theta & 1-\cos\theta \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$$

$$\begin{cases} (1) & (1+\cos\theta)\, a = -e^{-i\varphi}\sin\theta\, b \\ (2) & e^{i\varphi}\sin\theta\, a = -(1-\cos\theta)\, b \end{cases}$$

$$\begin{cases} (1) & 2\cos^2\frac{\theta}{2}\, a = -2\cos\frac{\theta}{2}\sin\frac{\theta}{2}\, e^{-i\varphi}\, b \\ (2) & 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\, e^{i\varphi}\, a = -2\sin^2\frac{\theta}{2}\, b \end{cases}$$

$$\begin{cases} (1) & \cos\frac{\theta}{2}\, a = -\sin\frac{\theta}{2}\, e^{-i\varphi}\, b \\ (2) & e^{i\varphi}\cos\frac{\theta}{2}\, a = -\sin\frac{\theta}{2}\, b \end{cases}$$

$\Rightarrow$ equations (1) and (2) are identical

$\Rightarrow$ choose $b = -e^{i\varphi}\cos\frac{\theta}{2}$ $\Rightarrow$ then $a = \sin\frac{\theta}{2}$

Clearly $|a|^2 + |b|^2 = \sin^2\frac{\theta}{2} + \cos^2\frac{\theta}{2} = 1$

$$\Rightarrow |-1\rangle = \sin\frac{\theta}{2}\,|+z\rangle - e^{i\varphi}\cos\frac{\theta}{2}\,|-z\rangle$$

In the textbook these states are denoted

$$|+1\rangle = |+n\rangle = \cos\frac{\theta}{2}\,|+z\rangle + e^{i\varphi}\sin\frac{\theta}{2}\,|-z\rangle \rightarrow \begin{pmatrix} \cos\frac{\theta}{2} \\ e^{i\varphi}\sin\frac{\theta}{2} \end{pmatrix}$$

$$|-1\rangle = |-n\rangle = \sin\frac{\theta}{2}\,|+z\rangle - e^{i\varphi}\cos\frac{\theta}{2}\,|-z\rangle \rightarrow \begin{pmatrix} \sin\frac{\theta}{2} \\ -e^{i\varphi}\cos\frac{\theta}{2} \end{pmatrix}$$

One can check that

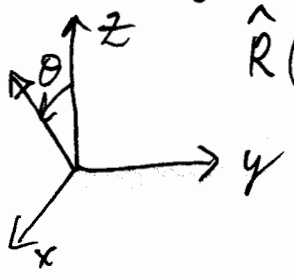
$$\langle -n|+n\rangle = \left(\sin\frac{\theta}{2}, -e^{-i\varphi}\cos\frac{\theta}{2}\right)\begin{pmatrix} \cos\frac{\theta}{2} \\ e^{i\varphi}\sin\frac{\theta}{2} \end{pmatrix} = \sin\frac{\theta}{2}\cos\frac{\theta}{2} - e^{-i\varphi}\cos\frac{\theta}{2}\, e^{i\varphi}\sin\frac{\theta}{2}$$

$$= 0.$$

<u>Solution by rotations:</u>

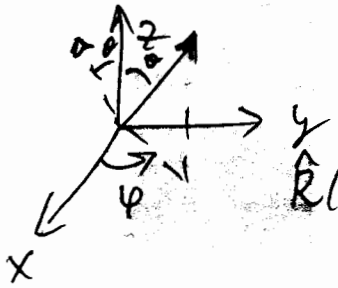
Start with $\hat{S}_z |+z\rangle = \frac{\hbar}{2} |+z\rangle$

and observe that we can rotate $|+z\rangle$ into $|+n\rangle$ state by (1) rotating about y-axis by angle $\theta$




$\hat{R}(\theta \vec{j})$

followed by (2) rotation about z-axis by angle $\varphi$




$\hat{R}(\varphi \vec{k})$

Thus $|+n\rangle = \hat{R}(\varphi \vec{k})\, \hat{R}(\theta \vec{j})\, |+z\rangle$ (the order of rotations is important!)

From lectures $\hat{R}(\varphi \vec{k}) \rightarrow \begin{pmatrix} e^{-i\varphi/2} & 0 \\ 0 & e^{i\varphi/2} \end{pmatrix}$

$$\hat{R}(\theta \vec{j}) \rightarrow \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$$

$$\Rightarrow |+n\rangle \rightarrow \begin{pmatrix} e^{-i\varphi/2} & 0 \\ 0 & e^{i\varphi/2} \end{pmatrix} \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} e^{-i\varphi/2} & 0 \\ 0 & e^{i\varphi/2} \end{pmatrix} \begin{pmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2} \end{pmatrix} =$$

$$= \begin{pmatrix} e^{-i\varphi/2} \cos\frac{\theta}{2} \\ e^{i\varphi/2} \sin\frac{\theta}{2} \end{pmatrix} = e^{-i\varphi/2} \begin{pmatrix} \cos\frac{\theta}{2} \\ e^{i\varphi} \sin\frac{\theta}{2} \end{pmatrix}.$$

But how do we know that this state is an eigenstate of $\hat{S}_n = \hat{S}\cdot\vec{n}$ ?

From $\hat{S}_z |+z\rangle = \frac{\hbar}{2} |+z\rangle$

we insert operator "unity" as

$$\hat{S}_z \underbrace{\hat{R}^{-1}(\theta_{\vec{j}})\, \hat{R}^{-1}(\varphi_{\vec{k}})\, \hat{R}(\varphi_{\vec{k}})\, \hat{R}(\theta_{\vec{j}})}_{\text{"1"}} |+z\rangle = \frac{\hbar}{2} |+z\rangle$$

But for rotation operators inverse = adjoint

$$\Rightarrow \hat{S}_z\, \hat{R}^{+}(\theta_{\vec{j}})\, \hat{R}^{+}(\varphi_{\vec{k}}) \left( \hat{R}(\varphi_{\vec{k}})\, \hat{R}(\theta_{\vec{j}}) |+z\rangle \right) = \frac{\hbar}{2} |+z\rangle$$

Multiply both sides by $\hat{R}(\varphi_{\vec{k}})\hat{R}(\theta_{\vec{j}})$ from left:

$$\Rightarrow \hat{R}(\varphi_{\vec{k}})\, \hat{R}(\theta_{\vec{j}})\, \hat{S}_z\, \hat{R}^{+}(\theta_{\vec{j}})\, \hat{R}^{+}(\varphi_{\vec{k}}) |+n\rangle = \frac{\hbar}{2} \hat{R}(\varphi_{\vec{k}})\hat{R}(\theta_{\vec{j}}) |+z\rangle$$

$$= \frac{\hbar}{2} |+n\rangle$$

$\Rightarrow$ by construction the rotated state

$|+n\rangle = \hat{R}(\varphi_{\vec{k}})\, \hat{R}(\theta_{\vec{j}}) |+z\rangle$ is an eigenstate

of rotated operator $\hat{A} = \hat{R}(\varphi_{\vec{k}})\, \hat{R}(\theta_{\vec{j}})\, \hat{S}_z\, \hat{R}^{+}(\theta_{\vec{j}})\, \hat{R}^{+}(\varphi_{\vec{k}})$

with eigenvalue $= \frac{\hbar}{2}$.

What is $\hat{A}$? $\rightarrow$ we shall find its matrix.

$\cancel{\hat{R}(\varphi_{\vec{k}}) = e^{-\frac{i}{\hbar}\hat{J}_z \varphi} = (\hat{J}_z = \frac{\hbar}{2}\sigma_z) = e^{-i\sigma_z \varphi/2} =}$

$\cancel{\text{(quantum Euler formula)}} \; \cancel{= \cos\frac{\varphi}{2} - i\sigma_z \sin\frac{\varphi}{2}}$

$$\hat{R}(\varphi\vec{k})\,\hat{R}(\theta\vec{j}) \rightarrow \begin{pmatrix} e^{-i\varphi/2} & 0 \\ 0 & e^{i\varphi/2} \end{pmatrix} \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix} =$$

$$= \begin{pmatrix} e^{-i\varphi/2}\cos\frac{\theta}{2} & -e^{-i\varphi/2}\sin\frac{\theta}{2} \\ e^{i\varphi/2}\sin\frac{\theta}{2} & e^{i\varphi/2}\cos\frac{\theta}{2} \end{pmatrix}$$

The adjoint of this matrix is

$$\left[\hat{R}(\varphi\vec{k})\,\hat{R}(\theta\vec{j})\right]^{\dagger} \rightarrow \begin{pmatrix} e^{i\varphi/2}\cos\frac{\theta}{2} & e^{-i\varphi/2}\sin\frac{\theta}{2} \\ -e^{i\varphi/2}\sin\frac{\theta}{2} & e^{-i\varphi/2}\cos\frac{\theta}{2} \end{pmatrix}$$

The matrix of $\hat{S}_z \rightarrow \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$\Rightarrow$ the matrix of $\hat{A}$ is

$$A \rightarrow \left(\frac{\hbar}{2}\right) \begin{pmatrix} e^{-i\varphi/2}\cos\frac{\theta}{2} & -e^{-i\varphi/2}\sin\frac{\theta}{2} \\ e^{i\varphi/2}\sin\frac{\theta}{2} & e^{i\varphi/2}\cos\frac{\theta}{2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} e^{i\varphi/2}\cos\frac{\theta}{2} & e^{-i\varphi/2}\sin\frac{\theta}{2} \\ -e^{i\varphi/2}\sin\frac{\theta}{2} & e^{-i\varphi/2}\cos\frac{\theta}{2} \end{pmatrix}$$

$$= \left(\frac{\hbar}{2}\right) \begin{pmatrix} e^{-i\varphi/2}\cos\frac{\theta}{2} & -e^{-i\varphi/2}\sin\frac{\theta}{2} \\ e^{i\varphi/2}\sin\frac{\theta}{2} & e^{i\varphi/2}\cos\frac{\theta}{2} \end{pmatrix} \begin{pmatrix} e^{i\varphi/2}\cos\frac{\theta}{2} & e^{-i\varphi/2}\sin\frac{\theta}{2} \\ e^{i\varphi/2}\sin\frac{\theta}{2} & -e^{-i\varphi/2}\cos\frac{\theta}{2} \end{pmatrix} =$$

$$= \left(\frac{\hbar}{2}\right) \begin{pmatrix} \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2} & e^{-i\varphi}\cos\frac{\theta}{2}\sin\frac{\theta}{2}\cdot 2 \\ e^{i\varphi}\sin\frac{\theta}{2}\cos\frac{\theta}{2}\cdot 2 & \sin^2\frac{\theta}{2} - \cos^2\frac{\theta}{2} \end{pmatrix} =$$

$$= \left(\frac{\hbar}{2}\right) \begin{pmatrix} \cos\theta & \sin\theta\, e^{-i\varphi} \\ e^{i\varphi}\sin\theta & -\cos\theta \end{pmatrix}$$

which is exactly the matrix representation of $\hat{S}\cdot\vec{n}$, see page 1 of this solution.

Thus we have shown that

$$\hat{S}\cdot\vec{n}\,|+n\rangle = \frac{\hbar}{2}\,|+n\rangle$$

and obtained $|+n\rangle = \hat{R}(\varphi\vec{k})\,\hat{R}(\theta\vec{j})\,|+z\rangle$ from a simple state $|+z\rangle \to \begin{pmatrix}1\\0\end{pmatrix}$.

Clearly the same rotation applied to $|-z\rangle \to \begin{pmatrix}0\\1\end{pmatrix}$ will give $|-n\rangle$.

Indeed $|-n\rangle = \begin{pmatrix} e^{-i\varphi/2} & 0 \\ 0 & e^{i\varphi/2}\end{pmatrix}\begin{pmatrix}\cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2}\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix} =$

$$= \begin{pmatrix} e^{-i\varphi/2} & 0 \\ 0 & e^{i\varphi/2}\end{pmatrix}\begin{pmatrix}-\sin\frac{\theta}{2} \\ \cos\frac{\theta}{2}\end{pmatrix} = \begin{pmatrix}-e^{-i\varphi/2}\sin\frac{\theta}{2} \\ e^{i\varphi/2}\cos\frac{\theta}{2}\end{pmatrix} =$$

$$= \left(-e^{-i\varphi/2}\right)\begin{pmatrix}\sin\theta/2 \\ -e^{i\varphi}\cos\frac{\theta}{2}\end{pmatrix}$$

$\Rightarrow |-n\rangle = \sin\frac{\theta}{2}\,|+z\rangle - e^{i\varphi}\cos\frac{\theta}{2}\,|-z\rangle$, as promised.

Finally, observe that since

$$\hat{S}_z\,|-z\rangle = -\frac{\hbar}{2}\,|-z\rangle$$

$$\underbrace{\hat{R}(\varphi\vec{k})\hat{R}(\theta\vec{j})\,\hat{S}_z\,\hat{R}^{+}(\theta\vec{j})\,\hat{R}^{+}(\varphi\vec{k})}_{\hat{S}\cdot\vec{n}}\;\underbrace{\hat{R}(\varphi\vec{k})\hat{R}(\theta\vec{j})|-z\rangle}_{|-n\rangle} = -\frac{\hbar}{2}\hat{R}(\varphi\vec{k})\hat{R}(\theta\vec{j})|-z\rangle$$

$\Rightarrow \hat{S}\cdot\vec{n}\,|-n\rangle = -\frac{\hbar}{2}|-n\rangle$ by construction.